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On the strong solutions of the 3D magneto-micropolar equations

F. W. Cruz^a and M. M. Novais^b

^aDepartamento de Matemática, Universidade Federal de Pernambuco – UFPE, Recife, PE, Brazil; ^bDepartamento de Matemática, Universidade Federal Rural de Pernambuco – UFRPE, Recife, PE, Brazil

ABSTRACT

This paper is dedicated to the study of micropolar fluids which is moving in the presence of a magnetic field in $\mathbb{R}^3$. We prove the global existence and uniqueness of strong solutions with initial data $(\mathbf{u}_0, \mathbf{w}_0, \mathbf{b}_0) \in H_\sigma^1(\mathbb{R}^3) \times H^1(\mathbb{R}^3) \times H_\sigma^1(\mathbb{R}^3)$ satisfying

$$\left(\|\mathbf{u}_0\|_{L^2}^2 + \|\mathbf{w}_0\|_{L^2}^2 + \|\mathbf{b}_0\|_{L^2}^2 \right) \times \left(\|\nabla \mathbf{u}_0\|_{L^2}^2 + \|\nabla \mathbf{w}_0\|_{L^2}^2 + \|\nabla \mathbf{b}_0\|_{L^2}^2 + \|\operatorname{curl} \mathbf{u}_0 - 2\mathbf{w}_0\|_{L^2}^2 \right)$$

being small enough.

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1. Introduction and main result

In this paper, we are concerned with the following model of incompressible magneto-micropolar fluids (see [1]):

$$\begin{aligned} \mathbf{u}_t + (\mathbf{u} \cdot \nabla) \mathbf{u} - (\mu + \chi) \Delta \mathbf{u} + \nabla \left(\Pi + \frac{|\mathbf{b}|^2}{2} \right) &= \chi \operatorname{curl} \mathbf{w} + (\mathbf{b} \cdot \nabla) \mathbf{b}, \quad \mathbf{x} \in \mathbb{R}^3, t > 0, \\ \mathbf{w}_t + (\mathbf{u} \cdot \nabla) \mathbf{w} - \gamma \Delta \mathbf{w} - \kappa \nabla (\operatorname{div} \mathbf{w}) + 2\chi \mathbf{w} &= \chi \operatorname{curl} \mathbf{u}, \quad \mathbf{x} \in \mathbb{R}^3, t > 0, \\ \mathbf{b}_t + (\mathbf{u} \cdot \nabla) \mathbf{b} - \nu \Delta \mathbf{b} &= (\mathbf{b} \cdot \nabla) \mathbf{u}, \quad \mathbf{x} \in \mathbb{R}^3, t > 0, \\ \operatorname{div} \mathbf{u} &= \operatorname{div} \mathbf{b} = 0, \quad \mathbf{x} \in \mathbb{R}^3, t > 0, \\ \mathbf{u}(\mathbf{x}, 0) &= \mathbf{u}_0(\mathbf{x}), \quad \mathbf{w}(\mathbf{x}, 0) = \mathbf{w}_0(\mathbf{x}), \quad \mathbf{b}(\mathbf{x}, 0) = \mathbf{b}_0(\mathbf{x}), \quad \mathbf{x} \in \mathbb{R}^3. \end{aligned} \quad (1)$$

In the above system, $\mathbf{u} = \mathbf{u}(\mathbf{x}, t) \in \mathbb{R}^3$ stands for the velocity vector, $\Pi = \Pi(\mathbf{x}, t) \in \mathbb{R}$ is the hydrostatic pressure, $\mathbf{w} = \mathbf{w}(\mathbf{x}, t) \in \mathbb{R}^3$ denotes the micro-rotational velocity and $\mathbf{b} = \mathbf{b}(\mathbf{x}, t) \in \mathbb{R}^3$ describes the magnetic field. The function $|\mathbf{b}|^2/2$ is the magnetic pressure. Thus, we denote by $p := \Pi + |\mathbf{b}|^2/2$ the total pressure of the fluid. The positive constants μ , χ , γ , κ and ν are associated with specific properties of the fluid; more specifically, μ is the kinematic viscosity, χ is the vortex viscosity, γ and κ are spin viscosities, and $1/\nu$ is the magnetic Reynolds number. Without loss of generality to our goals, we fix $\mu = \chi = 1/2$ and $\gamma = \kappa = \nu = 1$. We denote by $\mathbf{u}_0$, $\mathbf{w}_0$ and $\mathbf{b}_0$ the initial data for the linear velocity, the angular velocity and the magnetic field and we assume that $\operatorname{div} \mathbf{u}_0 = \operatorname{div} \mathbf{b}_0 = 0$.

Note that if $\chi = 0$ and $\mathbf{w} = \mathbf{b} = \mathbf{0}$, then system (1) is reduced to the classical Navier-Stokes equations (see, for instance, [2] and the references therein). On the other hand, observe that when the

magnetic field is absent, that is, $\mathbf{b} = \mathbf{0}$, the system (1) reduces to the micropolar fluid system, introduced by Eringen in [3] (see also [4]). Also notice that if $\chi = 0$ and $\mathbf{w} = \mathbf{0}$, the magneto-micropolar system (1) reduces to the MHD equations (see [5]).

Concerning the model considered in this paper, the local existence of strong solution, global existence of strong solutions for small initial data, and the global existence and uniqueness of weak solutions were studied by Rojas-Medar and collaborators in [6–8], respectively. In [9], the large time decay of system (1) on $\mathbb{R}^n$ ($n = 2$ and 3) was studied. In its turn, in [10], the authors applied the Fourier splitting method for the system (1) and obtained decay rates for the solution. The rates are the same as the ones for the Navier-Stokes equations. The global regularity of the 2D anisotropic magneto-micropolar fluid with mixed partial viscosity was studied in [11]. Very recently, Shang and Wu established in [12] the global regularity for the two-dimensional fractional magneto-micropolar system with the smallest amount of dissipation.

In [13], Wu established global regularity conditions to the MHD equations as follows: if $(\mathbf{u}, \mathbf{b})(\mathbf{x}, t)$ satisfies

$$\nabla \mathbf{u}, \nabla \mathbf{b} \in L^4(0, T; L^2(\mathbb{R}^3)),$$

then $\mathbf{u}$ and $\mathbf{b}$ remain smooth over $[0, T]$. Recently, Cruz proved in [14] the existence and uniqueness of global in time strong solutions for the 3D incompressible micropolar equations. In this work, we show the existence and uniqueness of global strong solutions to the system (1), therefore extending the results from [14] to the case of magneto-micropolar fluids. More precisely, our main result is the following (where $\mathbf{H}_\sigma^1(\mathbb{R}^3) := \{\mathbf{v} \in \mathbf{H}^1(\mathbb{R}^3) \mid \operatorname{div} \mathbf{v} = 0 \text{ in } \mathbb{R}^3\}$):

Theorem 1.1: *Given the initial data $(\mathbf{u}_0, \mathbf{w}_0, \mathbf{b}_0)$ satisfying*

$$(\mathbf{u}_0, \mathbf{w}_0, \mathbf{b}_0) \in \mathbf{H}_\sigma^1(\mathbb{R}^3) \times \mathbf{H}^1(\mathbb{R}^3) \times \mathbf{H}_\sigma^1(\mathbb{R}^3),$$

there exists a constant $\varepsilon_0 > 0$ independent of $\mathbf{u}_0, \mathbf{w}_0$ and $\mathbf{b}_0$ such that if

$$(\|\mathbf{u}_0\|_{L^2}^2 + \|\mathbf{w}_0\|_{L^2}^2 + \|\mathbf{b}_0\|_{L^2}^2)(\|\nabla \mathbf{u}_0\|_{L^2}^2 + \|\nabla \mathbf{w}_0\|_{L^2}^2 + \|\nabla \mathbf{b}_0\|_{L^2}^2 + \|\operatorname{curl} \mathbf{u}_0 - 2\mathbf{w}_0\|_{L^2}^2) \leq \varepsilon_0,$$

the IVP (1) has a unique global strong solution.

2. Preliminaries

2.1. The functional tool box

The function spaces $L^p(\mathbb{R}^3)$, $W^{m,p}(\mathbb{R}^3)$, $H^m(\mathbb{R}^3)$ and $L^p(0, T; \mathfrak{B})$ are defined as usual (here, $\mathfrak{B}$ is a Banach space and $0 < T \leq \infty$). Spaces of $\mathbb{R}^n$ valued functions, as well as their elements, are represented by bold face letters. The $L^2(\mathbb{R}^3)$ inner-product and norm are denoted by $\langle \cdot, \cdot \rangle$ and $\|\cdot\|$, respectively.

2.2. Key concepts

Definition 2.1 (Strong solution): Let $(\mathbf{u}_0, \mathbf{w}_0, \mathbf{b}_0) \in \mathbf{H}_\sigma^1(\mathbb{R}^3) \times \mathbf{H}^1(\mathbb{R}^3) \times \mathbf{H}_\sigma^1(\mathbb{R}^3)$. The triplet $(\mathbf{u}, \mathbf{w}, \mathbf{b})$ is called a strong solution of problem (1) on $(0, T)$ if

$$\mathbf{u}, \mathbf{w}, \mathbf{b} \in L^\infty(0, T; \mathbf{H}^1(\mathbb{R}^3)) \cap L^2(0, T; \mathbf{H}^2(\mathbb{R}^3)),$$

with $(\mathbf{u}, \mathbf{w}, \mathbf{b})$ satisfying Equations (1)₁–(1)₄ a.e. in $\mathbb{R}^3 \times (0, T)$, and the initial conditions (1)₅ in $\mathbf{H}_\sigma^1(\mathbb{R}^3) \times \mathbf{H}^1(\mathbb{R}^3) \times \mathbf{H}_\sigma^1(\mathbb{R}^3)$.

Definition 2.2 (Finite maximal time): A finite number $T^* > 0$ is called the finite maximal existence time (or finite blow-up time) for the strong solution of problem (1) if

$$\sup_{t \in [0, T]} (\|\nabla \mathbf{u}(t)\|^2 + \|\nabla \mathbf{w}(t)\|^2 + \|\nabla \mathbf{b}(t)\|^2) < \infty \quad \text{for all } T \in [0, T^*)$$

and

$$\lim_{t \uparrow T^*} \sup (\|\nabla \mathbf{u}(t)\|^2 + \|\nabla \mathbf{w}(t)\|^2 + \|\nabla \mathbf{b}(t)\|^2) = \infty.$$

In this case, we say that the solution $(\mathbf{u}, \mathbf{w}, \mathbf{b})$ blow-up in finite time.

3. Local solutions

The aim of this section is prove the following result of existence and uniqueness for local-in-time strong solutions.

Theorem 3.1: Suppose $(\mathbf{u}_0, \mathbf{w}_0, \mathbf{b}_0) \in \mathbf{H}_\sigma^1(\mathbb{R}^3) \times \mathbf{H}^1(\mathbb{R}^3) \times \mathbf{H}_\sigma^1(\mathbb{R}^3)$. There exists $T_0 \in (0, T]$ such that the problem (1) possesses exactly one strong solution $(\mathbf{u}, \mathbf{w}, \mathbf{b})$ defined in $\mathbb{R}^3 \times [0, T_0]$.

3.1. A priori estimates

In this subsection we get an *a priori* bound for solutions of problem (1) on $\mathbb{R}^3 \times (0, T)$. Let $\mathbf{z} := (\mathbf{u}, \mathbf{w}, \mathbf{b})$ be a sufficiently smooth solution. We have the following estimate in the $\mathbf{H}^1$ -norm.

Proposition 3.1: Let $(\mathbf{u}_0, \mathbf{w}_0, \mathbf{b}_0) \in \mathbf{H}_\sigma^1(\mathbb{R}^3) \times \mathbf{H}^1(\mathbb{R}^3) \times \mathbf{H}_\sigma^1(\mathbb{R}^3)$. Then there exists $T_0 \in (0, T]$ such that

$$\|\mathbf{u}(t)\|_{\mathbf{H}^1}^2 + \|\mathbf{w}(t)\|_{\mathbf{H}^1}^2 + \|\mathbf{b}(t)\|_{\mathbf{H}^1}^2 \leq C (\|\mathbf{u}_0\|_{\mathbf{H}^1}^2 + \|\mathbf{w}_0\|_{\mathbf{H}^1}^2 + \|\mathbf{b}_0\|_{\mathbf{H}^1}^2) + 1,$$

for all $t \in [0, T_0]$, where $C \in \mathbb{R}^+$ is independent of $\mathbf{u}_0, \mathbf{w}_0, \mathbf{b}_0$ and T_0 .

Proof: Multiplying Equation (1)₁ by $\mathbf{u}$, Equation (1)₂ by $\mathbf{w}$, and Equation (1)₃ by $\mathbf{b}$ in L^2 , and adding the resulting identities, one gets

$$\frac{1}{2} \frac{d}{dt} (\|\mathbf{z}\|^2) + \|\nabla \mathbf{z}\|^2 + \|\operatorname{div} \mathbf{w}\|^2 + \|\mathbf{w}\|^2 = \langle \operatorname{curl} \mathbf{u}, \mathbf{w} \rangle \leq \frac{1}{2} (\|\nabla \mathbf{u}\|^2 + \|\mathbf{w}\|^2),$$

and hence

$$\|\mathbf{z}(t)\|^2 + \int_0^t \|\nabla \mathbf{z}(\tau)\|^2 d\tau \leq \|\mathbf{z}_0\|^2. \quad (2)$$

Now, multiplying (dot product) the first, the second and the third equations in (1) by $\mathbf{u}_t, \mathbf{w}_t$ and $\mathbf{b}_t$, respectively, summing up and integrating the result on $\mathbb{R}^3$, one concludes that

$$\begin{aligned} & \|\mathbf{z}_t\|^2 + \frac{1}{2} \frac{d}{dt} \left(\frac{3}{4} \|\nabla \mathbf{u}\|^2 + \|\nabla \mathbf{w}\|^2 + \|\nabla \mathbf{b}\|^2 + \|\operatorname{div} \mathbf{w}\|^2 + \frac{1}{4} \|\operatorname{curl} \mathbf{u} - 2\mathbf{w}\|^2 \right) \\ &= \langle (\mathbf{u} \cdot \nabla) \mathbf{u}, -\mathbf{u}_t \rangle + \langle (\mathbf{u} \cdot \nabla) \mathbf{w}, -\mathbf{w}_t \rangle + \langle (\mathbf{u} \cdot \nabla) \mathbf{b}, -\mathbf{b}_t \rangle + \langle (\mathbf{b} \cdot \nabla) \mathbf{b}, \mathbf{u}_t \rangle + \langle (\mathbf{b} \cdot \nabla) \mathbf{u}, \mathbf{b}_t \rangle \\ &\leq \frac{1}{2} \|\mathbf{z}_t\|^2 + \|(\mathbf{u} \cdot \nabla) \mathbf{u}\|^2 + \frac{1}{2} \|(\mathbf{u} \cdot \nabla) \mathbf{w}\|^2 + \|(\mathbf{u} \cdot \nabla) \mathbf{b}\|^2 + \|(\mathbf{b} \cdot \nabla) \mathbf{b}\|^2 + \|(\mathbf{b} \cdot \nabla) \mathbf{u}\|^2, \end{aligned}$$

which shows that

$$\begin{aligned} \|z_t\|^2 + \frac{d}{dt} \left(\frac{3}{4} \|\nabla \mathbf{u}\|^2 + \|\nabla \mathbf{w}\|^2 + \|\nabla \mathbf{b}\|^2 + \|\operatorname{div} \mathbf{w}\|^2 + \frac{1}{4} \|\operatorname{curl} \mathbf{u} - 2\mathbf{w}\|^2 \right) \\ \leq 2 \left(\|(\mathbf{u} \cdot \nabla) \mathbf{u}\|^2 + \|(\mathbf{u} \cdot \nabla) \mathbf{w}\|^2 + \|(\mathbf{u} \cdot \nabla) \mathbf{b}\|^2 + \|(\mathbf{b} \cdot \nabla) \mathbf{b}\|^2 + \|(\mathbf{b} \cdot \nabla) \mathbf{u}\|^2 \right). \end{aligned} \quad (3)$$

We next take $-\Delta \mathbf{u}$, $-\Delta \mathbf{w}$ and $-\Delta \mathbf{b}$ as test functions in Equations (1)₁, (1)₂ and (1)₃, respectively. Since $\langle \operatorname{curl} \mathbf{u}, -\Delta \mathbf{w} \rangle = \langle \operatorname{curl} \mathbf{w}, -\Delta \mathbf{u} \rangle$, by adding the resulting equalities, one finds

$$\begin{aligned} \frac{1}{2} \frac{d}{dt} (\|\nabla \mathbf{z}\|^2) + \|\Delta \mathbf{z}\|^2 + \|\nabla(\operatorname{div} \mathbf{w})\|^2 + \|\nabla \mathbf{w}\|^2 \\ = \langle \operatorname{curl} \mathbf{w}, -\Delta \mathbf{u} \rangle + \langle (\mathbf{b} \cdot \nabla) \mathbf{b}, -\Delta \mathbf{u} \rangle + \langle (\mathbf{u} \cdot \nabla) \mathbf{u}, \Delta \mathbf{u} \rangle + \langle (\mathbf{u} \cdot \nabla) \mathbf{w}, \Delta \mathbf{w} \rangle \\ + \langle (\mathbf{b} \cdot \nabla) \mathbf{u}, -\Delta \mathbf{b} \rangle + \langle (\mathbf{u} \cdot \nabla) \mathbf{b}, \Delta \mathbf{b} \rangle \\ \leq \|\nabla \mathbf{w}\|^2 + \frac{5}{8} \|\Delta \mathbf{z}\|^2 + \frac{4}{3} (\|(\mathbf{u} \cdot \nabla) \mathbf{u}\|^2 + \|(\mathbf{b} \cdot \nabla) \mathbf{b}\|^2) \\ + \frac{2}{5} \|(\mathbf{u} \cdot \nabla) \mathbf{w}\|^2 + \frac{4}{5} (\|(\mathbf{u} \cdot \nabla) \mathbf{b}\|^2 + \|(\mathbf{b} \cdot \nabla) \mathbf{u}\|^2), \end{aligned}$$

and thus

$$\begin{aligned} \frac{d}{dt} (\|\nabla \mathbf{z}\|^2) + \frac{3}{4} \|\Delta \mathbf{z}\|^2 + 2 \|\nabla(\operatorname{div} \mathbf{w})\|^2 \\ \leq 3 (\|(\mathbf{u} \cdot \nabla) \mathbf{u}\|^2 + \|(\mathbf{u} \cdot \nabla) \mathbf{w}\|^2 + \|(\mathbf{u} \cdot \nabla) \mathbf{b}\|^2 + \|(\mathbf{b} \cdot \nabla) \mathbf{b}\|^2 + \|(\mathbf{b} \cdot \nabla) \mathbf{u}\|^2). \end{aligned} \quad (4)$$

Combining (3) with (4), one obtains

$$\begin{aligned} \frac{d}{dt} \left(\frac{7}{4} \|\nabla \mathbf{u}\|^2 + 2 \|\nabla \mathbf{w}\|^2 + 2 \|\nabla \mathbf{b}\|^2 + \|\operatorname{div} \mathbf{w}\|^2 + \frac{1}{4} \|\operatorname{curl} \mathbf{u} - 2\mathbf{w}\|^2 \right) \\ + \|\mathbf{z}_t\|^2 + \frac{3}{4} \|\Delta \mathbf{z}\|^2 + 2 \|\nabla(\operatorname{div} \mathbf{w})\|^2 \\ \leq 5 (\|(\mathbf{u} \cdot \nabla) \mathbf{u}\|^2 + \|(\mathbf{u} \cdot \nabla) \mathbf{w}\|^2 + \|(\mathbf{u} \cdot \nabla) \mathbf{b}\|^2 + \|(\mathbf{b} \cdot \nabla) \mathbf{b}\|^2 + \|(\mathbf{b} \cdot \nabla) \mathbf{u}\|^2). \end{aligned} \quad (5)$$

On the other hand, it follows from Gagliardo-Nirenberg and Sobolev's inequalities that

$$\begin{aligned} 5 \|(\mathbf{u} \cdot \nabla) \mathbf{u}\|^2 &\leq 5 \|\mathbf{u}\|_6^2 \|\nabla \mathbf{u}\|_3^2 \leq C \|\nabla \mathbf{u}\|^3 \|\Delta \mathbf{u}\| \leq \frac{3}{8} \|\Delta \mathbf{u}\|^2 + C \|\nabla \mathbf{u}\|^6, \\ 5 \|(\mathbf{u} \cdot \nabla) \mathbf{w}\|^2 &\leq C \|\nabla \mathbf{u}\|^2 \|\nabla \mathbf{w}\| \|\Delta \mathbf{w}\| \leq \frac{3}{8} \|\Delta \mathbf{w}\|^2 + C \|\nabla \mathbf{u}\|^4 \|\nabla \mathbf{w}\|^2, \\ 5 \|(\mathbf{u} \cdot \nabla) \mathbf{b}\|^2 &\leq C \|\nabla \mathbf{u}\|^2 \|\nabla \mathbf{b}\| \|\Delta \mathbf{b}\| \leq \frac{1}{8} \|\Delta \mathbf{b}\|^2 + C \|\nabla \mathbf{u}\|^4 \|\nabla \mathbf{b}\|^2, \\ 5 \|(\mathbf{b} \cdot \nabla) \mathbf{b}\|^2 &\leq C \|\nabla \mathbf{b}\|^3 \|\Delta \mathbf{b}\| \leq \frac{1}{8} \|\Delta \mathbf{b}\|^2 + C \|\nabla \mathbf{b}\|^6 \end{aligned}$$

and

$$5 \|(\mathbf{b} \cdot \nabla) \mathbf{u}\|^2 \leq 5 \|\mathbf{b}\|_\infty^2 \|\nabla \mathbf{u}\|^2 \leq C \|\nabla \mathbf{u}\|^2 \|\nabla \mathbf{b}\| \|\Delta \mathbf{b}\| \leq \frac{1}{8} \|\Delta \mathbf{b}\|^2 + C \|\nabla \mathbf{u}\|^4 \|\nabla \mathbf{b}\|^2.$$

Applying these estimates in (5) yields

$$\begin{aligned}
 & \frac{d}{dt} \left(\frac{7}{4} \|\nabla \mathbf{u}\|^2 + 2 \|\nabla \mathbf{w}\|^2 + 2 \|\nabla \mathbf{b}\|^2 + \|\operatorname{div} \mathbf{w}\|^2 + \frac{1}{4} \|\operatorname{curl} \mathbf{u} - 2\mathbf{w}\|^2 \right) \\
 & \quad + \|\mathbf{z}_t\|^2 + \frac{3}{8} \|\Delta \mathbf{z}\|^2 + 2 \|\nabla(\operatorname{div} \mathbf{w})\|^2 \\
 & \leq C \|\nabla \mathbf{u}\|^4 \|\nabla \mathbf{z}\|^2 + C (\|\nabla \mathbf{u}\|^4 + \|\nabla \mathbf{b}\|^4) \|\nabla \mathbf{b}\|^2 \\
 & \leq C (\|\nabla \mathbf{u}\|^4 + \|\nabla \mathbf{b}\|^4) \|\nabla \mathbf{z}\|^2.
 \end{aligned} \tag{6}$$

Then we infer from [15, Lemma 6, p. 1098] and (6) that there exists $T_0 \in (0, T]$ such that, for any $t \leq T_0$,

$$\|\nabla \mathbf{z}(t)\|^2 \leq \|\nabla \mathbf{u}_0\|^2 + \frac{12}{7} \|\nabla \mathbf{w}_0\|^2 + \frac{8}{7} \|\nabla \mathbf{b}_0\|^2 + \frac{1}{7} \|\operatorname{curl} \mathbf{u}_0 - 2\mathbf{w}_0\|^2 + 1,$$

which together with bound (2) implies that

$$\|\mathbf{z}(t)\|_{H^1}^2 \leq C \|\mathbf{z}_0\|_{H^1}^2 + 1,$$

for all $t \in [0, T_0]$. This completes the proof of proposition. ■

Remark 3.1: We note that $\|\nabla \mathbf{b}\|^4$ appears on the rhs of (6) due to the term $\|(\mathbf{b} \cdot \nabla) \mathbf{b}\|^2$. Hence, through energy estimate, it is not possible to eliminate it.

3.2. Idea of proof of Theorem 3.1

The local existence of strong solutions to system (1) is proved through a spectral Galerkin method based in the estimates established in Proposition 3.1 (see e.g. [7]). The uniqueness part of Theorem 3.1 is proved in a completely similar way as in [14, Theorem 3.1, p. 206]. We omit it here the details. This finishes the proof.

4. Global solutions: Proof of main theorem

From now on, we consider

$$\begin{aligned}
 C_0 &:= \|\mathbf{u}_0\|^2 + \|\mathbf{w}_0\|^2 + \|\mathbf{b}_0\|^2 \quad \text{and} \\
 K_0 &:= 2 (\|\nabla \mathbf{u}_0\|^2 + \|\nabla \mathbf{w}_0\|^2 + \|\nabla \mathbf{b}_0\|^2 + \|\operatorname{curl} \mathbf{u}_0 - 2\mathbf{w}_0\|^2).
 \end{aligned}$$

Furthermore, let $\mathbf{z} := (\mathbf{u}, \mathbf{w}, \mathbf{b})$ be the strong solution of problem (1) on $\mathbb{R}^3 \times (0, T)$.

Lemma 4.1: *There exists a constant $C > 0$ independent of T , $\mathbf{u}_0$, $\mathbf{w}_0$ and $\mathbf{b}_0$, such that for all $t \in (0, T)$, there holds*

$$\sup_{\tau \in [0, t]} (\|\nabla \mathbf{u}(\tau)\|^2 + \|\nabla \mathbf{w}(\tau)\|^2 + \|\nabla \mathbf{b}(\tau)\|^2) \leq K_0 + CC_0 \sup_{\tau \in [0, t]} (\|\nabla \mathbf{u}(\tau)\|^4 + \|\nabla \mathbf{b}(\tau)\|^4). \tag{7}$$

Proof: It follows directly from inequality (2) that

$$\|\mathbf{u}(t)\|^2 + \|\mathbf{w}(t)\|^2 + \|\mathbf{b}(t)\|^2 + \int_0^t (\|\nabla \mathbf{u}(\tau)\|^2 + \|\nabla \mathbf{w}(\tau)\|^2 + \|\nabla \mathbf{b}(\tau)\|^2) d\tau \leq C_0, \quad (8)$$

for every $t \in [0, T]$. Integrating bound (6) over time variable and using the energy inequality (8), we get

$$\begin{aligned} & \|\nabla \mathbf{z}(t)\|^2 + \frac{4}{7} \int_0^t \|\mathbf{z}_t(\tau)\|^2 d\tau + \frac{3}{14} \int_0^t \|\Delta \mathbf{z}(\tau)\|^2 d\tau \\ & \leq \|\nabla \mathbf{u}_0\|^2 + \frac{12}{7} \|\nabla \mathbf{w}_0\|^2 + \frac{8}{7} \|\nabla \mathbf{b}_0\|^2 + \frac{1}{7} \|\operatorname{curl} \mathbf{u}_0 - 2\mathbf{w}_0\|^2 \\ & \quad + C \int_0^t (\|\nabla \mathbf{u}(\tau)\|^4 + \|\nabla \mathbf{b}(\tau)\|^4) \|\nabla \mathbf{z}(\tau)\|^2 d\tau \\ & \leq K_0 + C \sup_{\tau \in [0, t]} (\|\nabla \mathbf{u}(\tau)\|^4 + \|\nabla \mathbf{b}(\tau)\|^4) \int_0^t \|\nabla \mathbf{z}(\tau)\|^2 d\tau \\ & \leq K_0 + CC_0 \sup_{\tau \in [0, t]} (\|\nabla \mathbf{u}(\tau)\|^4 + \|\nabla \mathbf{b}(\tau)\|^4). \end{aligned}$$

This proves estimate (7). ■

Lemma 4.2: *There exists a positive constant ε_0 independent of T , $\mathbf{u}_0$, $\mathbf{w}_0$ and $\mathbf{b}_0$, such that*

$$\sup_{t \in [0, T]} (\|\nabla \mathbf{u}(t)\|^2 + \|\nabla \mathbf{w}(t)\|^2 + \|\nabla \mathbf{b}(t)\|^2) \leq 2K_0, \quad (9)$$

provided that

$$C_0 K_0 \leq 2\varepsilon_0. \quad (10)$$

Proof: We define

$$\mathcal{E}(t) := \sup_{\tau \in [0, t]} (\|\nabla \mathbf{u}(\tau)\|^2 + \|\nabla \mathbf{w}(\tau)\|^2 + \|\nabla \mathbf{b}(\tau)\|^2).$$

Observe that

$$\mathcal{E}(0) = \|\nabla \mathbf{u}_0\|^2 + \|\nabla \mathbf{w}_0\|^2 + \|\nabla \mathbf{b}_0\|^2 \leq K_0 < 4K_0. \quad (11)$$

By Lemma 4.1, one obtains

$$\mathcal{E}(t) \leq K_0 + MC_0 \mathcal{E}(t)^2, \quad (12)$$

where $M \in \mathbb{R}^+$ is a constant independent of T , $\mathbf{u}_0$, $\mathbf{w}_0$ and $\mathbf{b}_0$. Next, assume that

$$MC_0 K_0 \leq \frac{1}{8}, \quad (13)$$

and let

$$T_\star := \sup \{t \in [0, T] \mid \mathcal{E}(\tau) \leq 4K_0, \forall \tau \in [0, t]\}. \quad (14)$$

By the continuity of $\mathcal{E}$ on $[0, T]$, we conclude that the maximum is attained. We claim that $T_\star = T$. In fact, suppose, for the purpose of contradiction, that $T_\star \neq T$. From (11) we infer that $T_\star \in (0, T)$.

Then, it follows from (12)–(14) that

$$\mathcal{E}(T_\star) \leq K_0 + MC_0 \mathcal{E}^2(T_\star) \leq K_0 + 4MC_0 K_0 \mathcal{E}(T_\star) \leq K_0 + \frac{1}{2} \mathcal{E}(T_\star).$$

Consequently,

$$\mathcal{E}(T_\star) \leq 2K_0 < 4K_0.$$

This contradicts the definition of $T_\star$ in (14). Thus, $T_\star = T$. Lastly, choosing $\varepsilon_0 \equiv (1/16M)$, we find

$$\mathcal{E}(t) \leq 2K_0,$$

for all $t \in [0, T]$, provided that (10) holds. So, we complete the proof of Lemma 4.2. ■

Lemma 4.3 (Blow-up criterion): Suppose $(\mathbf{u}_0, \mathbf{w}_0, \mathbf{b}_0) \in \mathbf{H}_\sigma^1(\mathbb{R}^3) \times \mathbf{H}^1(\mathbb{R}^3) \times \mathbf{H}_\sigma^1(\mathbb{R}^3)$. Then T^* is the finite blow-up time of $(\mathbf{u}, \mathbf{w}, \mathbf{b})$ if and only if T^* is the least time such that

$$\|(\nabla \mathbf{u}, \nabla \mathbf{b})\|_{L^4(0, T^*; L^2(\mathbb{R}^3))} = \infty.$$

Proof: Using Gronwall inequality to (6), one gets

$$\mathcal{E}(T) \leq K_0 \times \exp \left(C \int_0^T (\|\nabla \mathbf{u}(t)\|^4 + \|\nabla \mathbf{b}(t)\|^4) dt \right), \quad \forall T \in [0, T^*].$$

The proof of Lemma 4.3 is finished. ■

Proof of Theorem 1.1

Suppose that $(\mathbf{u}_0, \mathbf{w}_0, \mathbf{b}_0) \in \mathbf{H}_\sigma^1(\mathbb{R}^3) \times \mathbf{H}^1(\mathbb{R}^3) \times \mathbf{H}_\sigma^1(\mathbb{R}^3)$ and that

$$(\|\mathbf{u}_0\|^2 + \|\mathbf{w}_0\|^2 + \|\mathbf{b}_0\|^2) (\|\nabla \mathbf{u}_0\|^2 + \|\nabla \mathbf{w}_0\|^2 + \|\nabla \mathbf{b}_0\|^2 + \|\operatorname{curl} \mathbf{u}_0 - 2\mathbf{w}_0\|^2) \leq \varepsilon_0,$$

where $\varepsilon_0 > 0$ is the constant stated in Lemma 4.2. Let $(\mathbf{u}, \mathbf{w}, \mathbf{b})$ be the unique local strong solution of problem (1) given by Theorem 3.1. Moreover, let $[0, T^*)$ be the maximal interval of existence to the strong solution of problem (1). We will prove that $T^* = \infty$. Suppose, by contradiction, that $T^* < \infty$. By Lemma 4.3, we have that

$$\int_0^{T^*} (\|\nabla \mathbf{u}(t)\|^4 + \|\nabla \mathbf{b}(t)\|^4) dt = \infty. \quad (15)$$

In addition, from Lemma 4.2, we infer

$$\sup_{t \in [0, T]} (\|\nabla \mathbf{u}(t)\|^2 + \|\nabla \mathbf{w}(t)\|^2 + \|\nabla \mathbf{b}(t)\|^2) \leq 2K_0, \quad \text{for every } T \in [0, T^*].$$

Consequently,

$$\int_0^{T^*} (\|\nabla \mathbf{u}(t)\|^4 + \|\nabla \mathbf{b}(t)\|^4) dt \leq 4K_0^2 T^* < \infty.$$

This contradicts (15). Then, $T^* = \infty$ and the solution is global in time. This ends the proof.

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